

New antenna design

Ready during fall 2026

Revision PA3



IFX radar (ver 1 ~Large aperture)

Concept

Size

- 100x60 mm
- Based on Golden Devices Antenna

Front-end

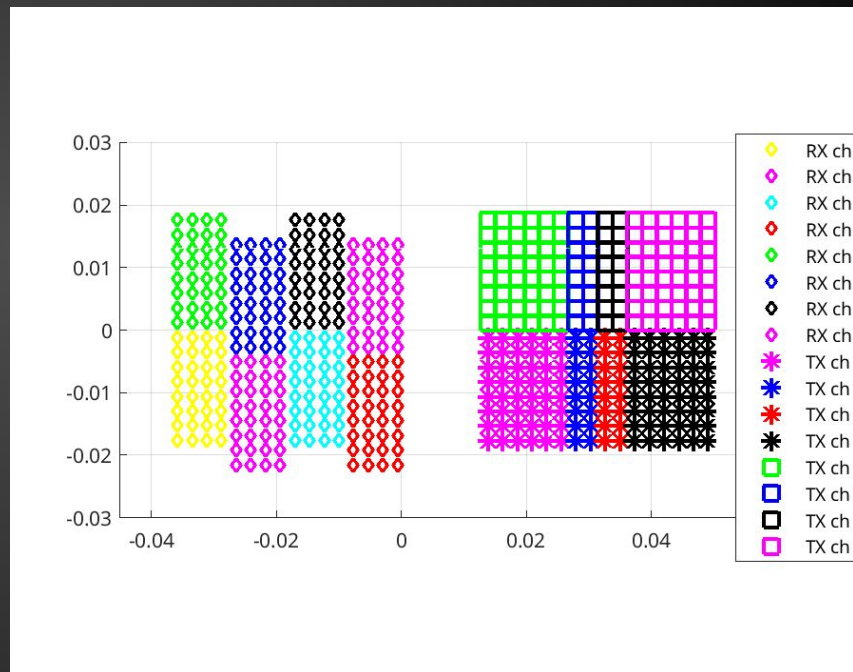
- Antenna layout =>
- 8 RX and 8 TX
 - 8 TX coherently
 - Staggered at RX of lambda pitch=>
- Covers 1 GHz system bandwidth
- Center frequency 76.5 GHz

Chipset

- CTRX from IFX
I2C comm+ Ethernet



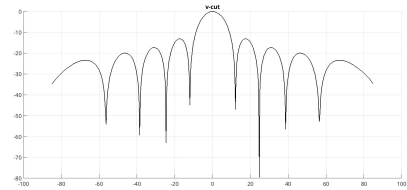
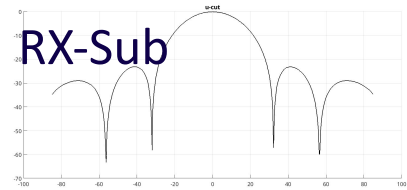
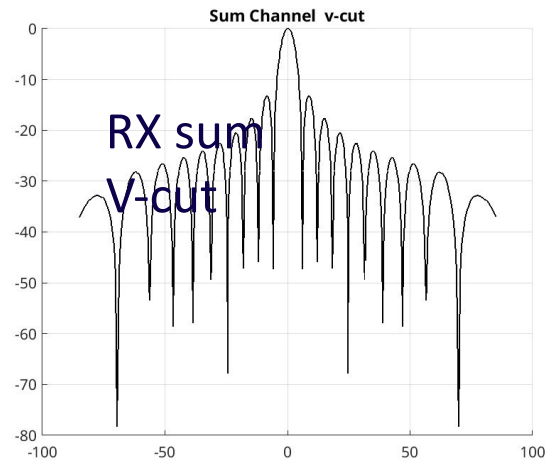
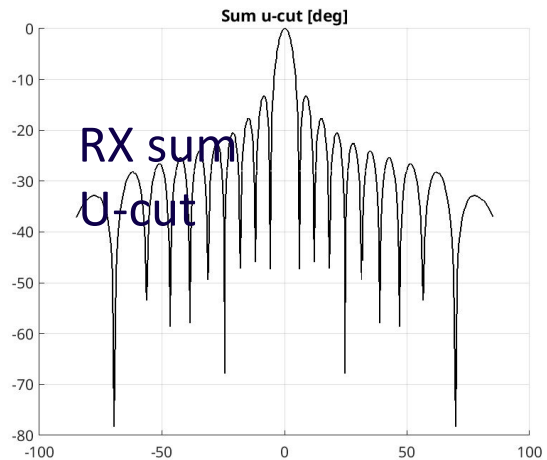
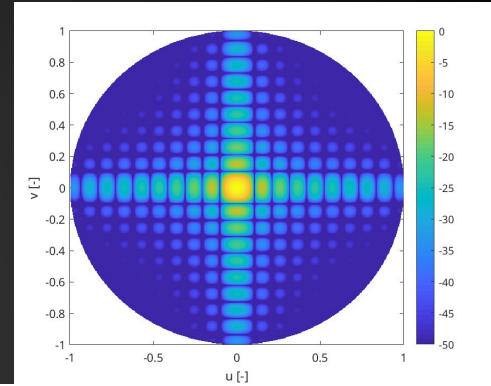
2016-09-07





Antenna Pattern Characteristics (RX)

- RX beam Characteristics
 - Without phase & amplitude taper:
 - Directivity sum beam RX >28 dBi
 - TX sum Beam (one way, 3dB):
 - Az and El: $\sim \pm 3$ deg
- Directivity RX sub channel: ~ 21 dBi

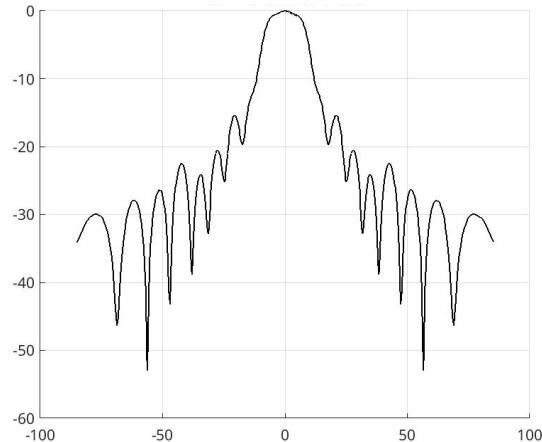


Antenna Pattern Characteristics (TX)

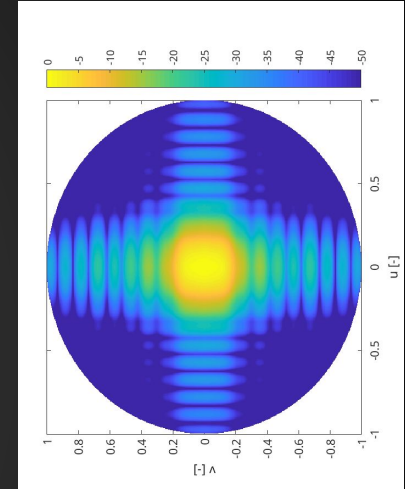
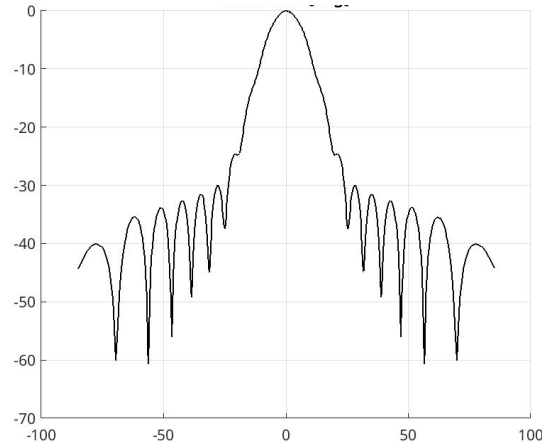
TX Beam Characteristics

- With phase and amplitude taper and Considering feeding of waveguide and power division => smaller center channels
- With phase and amplitude taper:
(Taylor 35dB taper in aperture => no power loss)
 - Directivity sum beam TX sum Beam (one way, 3dB): ~23.5 dBi
 - Az and El (3dB single way): $> \pm 6.5$ deg

U-cut



V-cut





- **Total Size**
 - ~100x60x10mm (Weight: TBD)
- **Frequency coverage (system bandwidth)**
 - 76-77GHz
- **Polarization**
 - Vertical
- **Directivity & Beamwidth**
 - TX: >23 dBi @ boresight having a sum beamwidth (3dB one way) of $\geq \pm 6.5$ degrees (3dB) in azimuth and elevation, respectively.
 - RX: >28 dBi @ boresight having a sum beamwidth (3dB one way) of $\geq \pm 3$ degrees (3dB) in azimuth and elevation, respectively.
 - Subapertures shall have a RX-directivity > 21 dBi and having a beamwidth (3dB one way) of $\geq \pm 10$ degrees and $\geq \pm 5$ degrees in azimuth and elevation, respectively
- **Tapering**
 - TX:
 - ~2D Taylor 35dB (Peak SLL 35 dB in az and el) partitioned over as above aperture
 - Beam widening using quadratic phase taper
 - RX: uniform taper (Peak SLL 13 dB in az and el) along azimuth of each of the RX channels
- **Isolation**
 - RX-TX: 35 dB
 - TX-TX: 15 dB
 - RX-RX: 15 dB
- **Chipset layout and connection: According to IFX CTRX8188F**
 - Return loss ≤ -20 dB
- **Radiation efficiency >80%**
- **Average amplitude mismatch between RX-subapertures $\leq \pm 1$ dB**



Concept

Size

- 80x70 mm
- Based on Golden Devices Antenna

Front-end

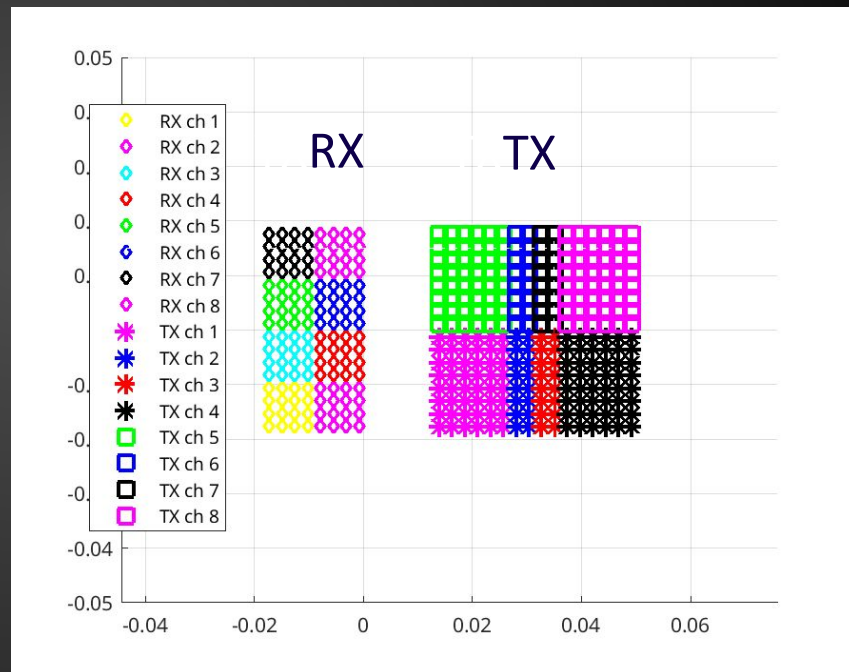
- Antenna layout =>
- 8 RX and 8 TX
 - 8 TX coherently
- Covers 1 GHz system bandwidth
- Center frequency 76.5 GHz

Chipset

- CTRX from IFX
I2C comm+ Ethernet



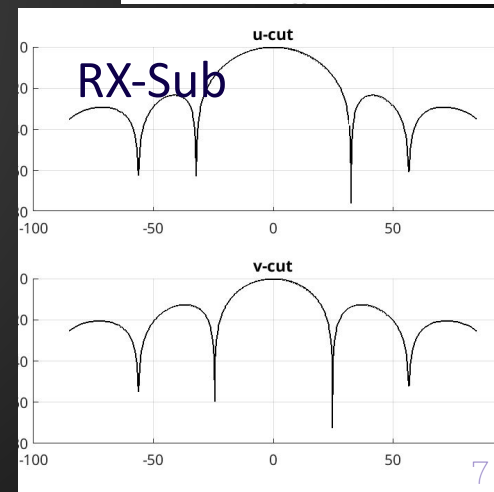
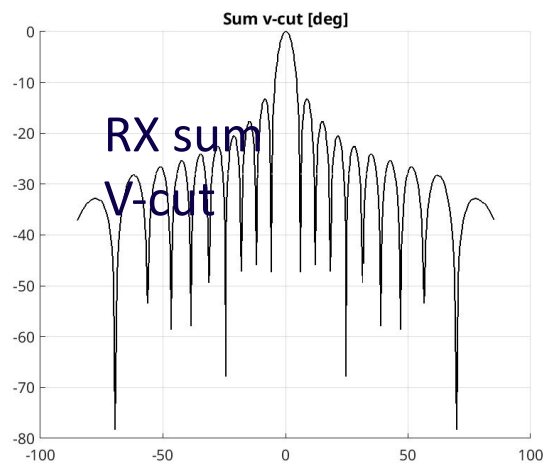
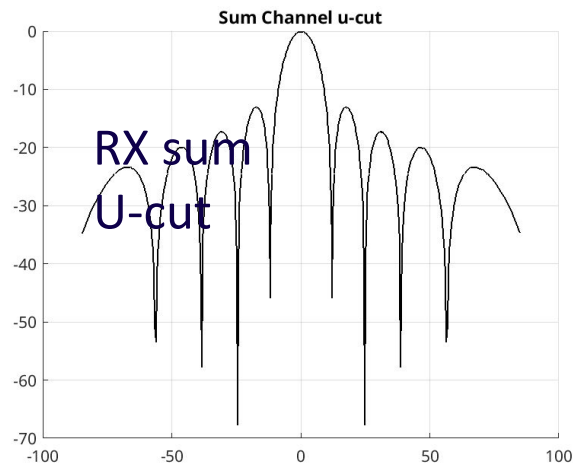
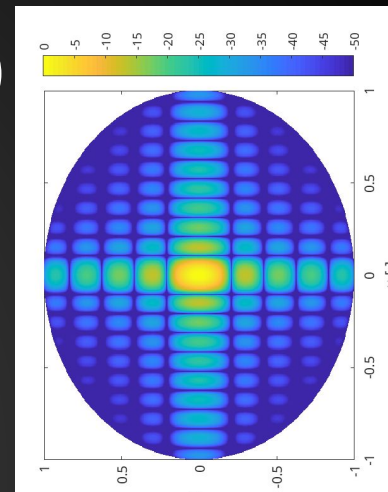
2016-09-07





Antenna Pattern Characteristics (RX)

- RX beam Characteristics
 - Without phase & amplitude taper:
 - Directivity sum beam RX >28 dBi
 - TX sum Beam (one way, 3dB):
 - Az and El: $\sim \pm 6$ deg and ± 3 deg, respectively.
- Directivity RX sub channel: ~ 21 dBi

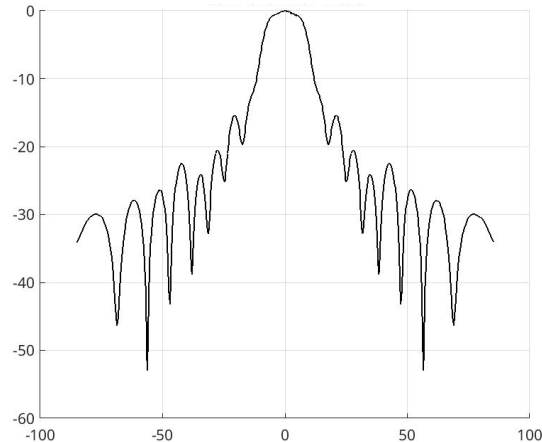


Antenna Pattern Characteristics (TX)

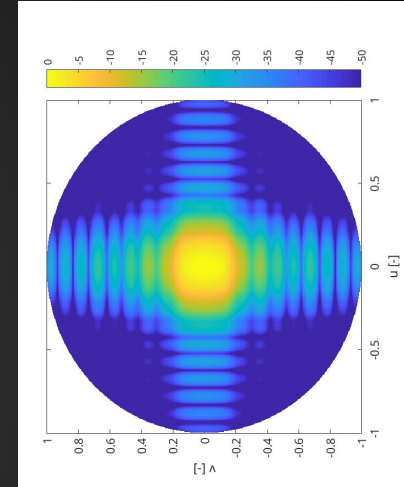
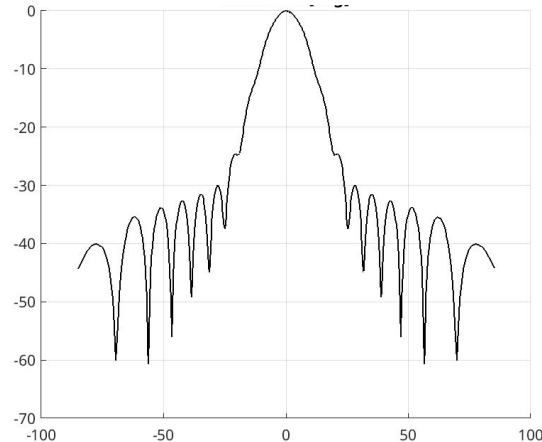
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 - Az and El (3dB single way): $> \pm 6.5$ deg

U-cut



V-cut





- **Total Size**
 - ~80x70x10mm (Weight: TBD)
- **Frequency coverage (system bandwidth)**
 - 76-77GHz
- **Polarization**
 - Vertical
- **Directivity & Beamwidth**
 - TX: >23 dBi @ boresight having a sum beamwidth (3dB one way) of $\geq \pm 3$ degrees (3dB) degrees in azimuth and elevation, respectively.
 - RX: >25 dBi @ boresight having a sum beamwidth (3dB one way) of $\geq \pm 6$ and $\geq \pm 3$ degrees (3dB) degrees in azimuth and elevation, respectively.
 - Subapertures shall have a RX-directivity > 18 dBi and having a beamwidth (3dB one way) of $\geq \pm 10$ degrees and $\geq \pm 10$ degrees in azimuth and elevation, respectively
- **Tapering**
 - TX:
 - 2D Taylor
 - Channel 2 and 3 (central): uniform taper (Peak SLL 13 dB in az) along azimuth
 - RX: uniform taper (Peak SLL 13 dB in az and el) along azimuth of each of the RX channels
- **Isolation**
 - RX-TX: 35 dB
 - TX-TX: 15 dB
 - RX-RX: 15 dB
- **Chipset layout and connection: According to IFX CTRX8188F**
 - Return loss <-20dB
- **Radiation efficiency >80%**
- **Average amplitude mismatch between RX-subapertures < ± 1 dB**